Case Study 6: Student Timetable Conflict Detection

Aim: To develop a Python program that compares course schedules using strings and tuples, detects overlapping class periods, identifies unavailable classrooms, and validates a proposed student timetable.

1. Problem Statement

A university conducts multiple courses in different classrooms and time slots. Sometimes two courses may be scheduled at the same time, or two classes may be assigned to the same room.

The program should:

1. Accept course names and course schedules.

2. Store schedules using tuples.

3. Compare the time periods of different courses.

4. Detect overlapping class timings.

5. Identify unavailable classrooms.

6. Validate a proposed timetable.

7. Display the final timetable status.

8. Assumed Conditions

| Parameter | Details |
|---|---|
| Working days | Monday to Saturday |
| Class timings | 9 AM to 5 PM |
| Time format | 24-hour format |
| Room availability | One class per room at a time |
| Student rule | A student cannot attend two classes simultaneously |

3. Algorithm

4. Start.

5. Enter the number of courses.

6. Read course name, day, start time, end time, and room number.

7. Store each course as a tuple.

8. Compare every pair of courses.

9. Check whether their days and time periods overlap.

10. If the courses have the same room, identify a room conflict.

11. If the courses belong to the same student timetable, identify a student conflict.

12. Display all conflicts.

13. Display whether the proposed timetable is valid.

14. Stop.

15. Pseudocode

```
START

INPUT number_of_courses

CREATE empty timetable list

FOR each course

    INPUT course_name
    INPUT day
    INPUT start_time
    INPUT end_time
    INPUT room_number

    STORE course details as tuple
```

```
END FOR

SET conflict = FALSE

FOR each pair of courses

    IF both courses have the same day

        IF start1 < end2 AND start2 < end1

            IF room1 = room2
                DISPLAY Room conflict
                conflict = TRUE
            END IF

            DISPLAY Time overlap
            conflict = TRUE

        END IF

    END IF

END FOR

IF conflict = FALSE
    DISPLAY Timetable is valid
```

ELSE

DISPLAY Timetable has conflicts

END IF

STOP

5. Python Program

Complete Python solution

# Student Timetable Conflict Detectiontimetable = []n = int(input("Enter number of courses: "))# Collect course detailsfor i in range(n): print("\nCourse", i + 1) course = input("Enter course name: ") day = input("Enter day: ").lower() start = int(input("Enter start time (24-hour format): ")) end = int(input("Enter end time (24-hour format): ")) room = input("Enter room number: ") if start < 9 or end > 17 or start >= end: print("Invalid time! Course skipped.") continue # Store course details as tuple timetable.append((course, day, start, end, room))# Check timetable conflictsconflict_found = Falseprint("\n====== TIMETABLE CONFLICT REPORT ======")for i in range(len(timetable)):

6. Sample Output

Enter number of courses: 3

Course 1

Enter course name: Python

Enter day: Monday

Enter start time: 9

Enter end time: 11

Enter room number: R101

Course 2

Enter course name: Mathematics

Enter day: Monday

Enter start time: 10

Enter end time: 12

Enter room number: R102

Course 3

Enter course name: English

Enter day: Tuesday

Enter start time: 9

Enter end time: 10

Enter room number: R103

====== TIMETABLE CONFLICT REPORT ======

Conflict detected:

Python and Mathematics

Student timetable conflict.

====== PROPOSED TIMETABLE ======

Python monday 9:00 11:00 R101

Mathematics monday 10:00 12:00 R102

English tuesday 9:00 10:00 R103

====== FINAL RESULT ======

Timetable is NOT VALID.

Please change the conflicting schedule.

7. Flowchart

#chatgpt-mermaid-_r_28l_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_28l_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_28l_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_28l_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_28l_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_28l_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_28l_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_28l_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_28l_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_28l_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_28l_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_28l_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_28l_ p{margin:0;}#chatgpt-mermaid-_r_28l_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_28l_ .label text,#chatgpt-mermaid-_r_28l_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .node rect,#chatgpt-mermaid-_r_28l_

.node circle,#chatgpt-mermaid-_r_28l_ .node ellipse,#chatgpt-mermaid-_r_28l_ .node polygon,#chatgpt-mermaid-_r_28l_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .rough-node .label text,#chatgpt-mermaid-_r_28l_ .node .label text,#chatgpt-mermaid-_r_28l_ .image-shape .label,#chatgpt-mermaid-_r_28l_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_28l_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .rough-node .label,#chatgpt-mermaid-_r_28l_ .node .label,#chatgpt-mermaid-_r_28l_ .image-shape .label,#chatgpt-mermaid-_r_28l_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_28l_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_28l_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_28l_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_28l_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_28l_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_28l_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_28l_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_28l_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_28l_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_28l_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_28l_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_28l_ .icon-shape,#chatgpt-mermaid-_r_28l_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_28l_ .icon-shape p,#chatgpt-mermaid-_r_28l_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_28l_ .icon-shape .label rect,#chatgpt-mermaid-_r_28l_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_28l_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_28l_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_28l_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_28l_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_28l_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_28l_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_28l_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_28l_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_28l_ [data-

look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_28l_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ [data-look="neo"].node .outer-path{filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_28l_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_28l_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_28l_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_28l_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_28l_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_28l_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_28l_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_28l_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_28l_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_28l_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_28l_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_28l_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_28l_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .node rect,#chatgpt-mermaid-_r_28l_ .node circle,#chatgpt-mermaid-_r_28l_ .node ellipse,#chatgpt-mermaid-_r_28l_ .node polygon,#chatgpt-mermaid-_r_28l_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_28l_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_28l_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_28l_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_28l_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_28l_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter number of coursesEnter course detailsStore details in tuplesSelect two coursesSame day?Check next pairTime periods overlap?Report student conflictSame room?Report room conflictMore pairs?Any conflicts?Timetable invalidTimetable validStopNoYesNoYesYesNoYesNoYesNo

8. Concepts Used from Syllabus

| Syllabus concept | Application |
| --- | --- |
| Variables | Course name, start time |
| Data types | Integer and string |
| Operators | Comparison operators |
| Conditional statements | Conflict detection |
| Loops | Comparing multiple courses |
| Strings | Course names and weekdays |
| Lists | Storing timetable records |
| Tuples | Representing course schedules |
| Tuple unpacking | Extracting course details |
| Algorithms | Timetable validation |

9. Conclusion

The Student Timetable Conflict Detection program checks course schedules and identifies overlapping periods and classroom conflicts. It validates the proposed timetable and helps universities avoid scheduling problems. This program demonstrates the practical application of Python lists, tuples, strings, loops, and conditional statements.

Note: This simplified program assumes every course belongs to the same student's timetable. It checks room conflicts between the entered courses but does not check against a separate room-booking database.